

DBMS LAB PROGRAM

1. DATABASE AND TABLE CREATION

```
1 ● show databases;
2 ● create database test;
3 use test;
4
5 ● ⊖ create table person(
6     driver_id varchar(10) primary key,
7     name varchar(20),
8     adress varchar(30));
9 ● desc person;
10
11 ● ⊖ create table car(
12     reg_num varchar(10) primary key,
13     model varchar(10),
14     year int);
15 ● desc car;
16
17 ● ⊖ create table accident(
18     report_num int primary key,
19     accident_date date,
20     location varchar(20) );
21 ● desc accident;
22 --
23 ● ⊖ create table owns(driver_id varchar(10), reg_num varchar(10),
24     primary key(driver_id, reg_num),
25     foreign key(driver_id) references person(driver_id),
26     foreign key(reg_num) references car(reg_num));
27 ● desc owns;
28
29 ● ⊖ create table participated(driver_id varchar(10), reg_num varchar(10),
30     report_num int, damage_amount int,
31     primary key(driver_id, reg_num, report_num),
32     foreign key(driver_id) references person(driver_id),
33     foreign key(reg_num) references car(reg_num),
34     foreign key(report_num) references accident(report_num));
35 ● desc participated;
36 --
```

2.INSERTING VALUES INTO TABLE

```
insert into person values("A01","richard","srinivas nagar"),
("A02","pradeep","rajaaji nagar"),
("A03","smith","ashok nagar"),
("A04","venu","nk colony"),
("A05","john","hanuman nagar");
```

```
insert into car values("KA052250","indica",1990),
("KA052251","toyota",1957),
("KA052252","honda",1998),
("KA052253","audi",2008),
("KA052254","lambo",2005);
```

```
insert into owns values("A01","KA052250"),
("A02","KA052251"),
("A03","KA052252"),
("A04","KA052253"),
("A05","KA052254");
```

```
INSERT INTO accident VALUES(11, '2003-01-01', 'Mysore Road'),
(12, '2004-02-02', 'South end Circle'),
(13, '2003-01-21', 'Bull temple Road'),
(14, '2008-02-17', 'Mysore Road'),
(15, '2005-03-04', 'Kanakpura Road');
```

```
INSERT INTO participated VALUES('A01', 'KA052250', 11, 10000),
('A02', 'KA052251', 12, 20000),
('A03', 'KA052252', 13, 25000),
('A04', 'KA052253', 14, 30000),
('A05', 'KA052254', 15, 50000);
```

VIEWING THE TABLES

```
select * from person;
select * from car;
select * from owns;
select*from accident;
select*from participated;
```

TABLES OUTPUT

1. PERSON

Result Grid			
	driver_id	name	adress
▶	A01	richard	srinivas nagar
	A02	pradeep	rajaji nagar
	A03	smith	ashok nagar
	A04	venu	nk colony
	A05	john	hanuman nagar
*	NULL	NULL	NULL

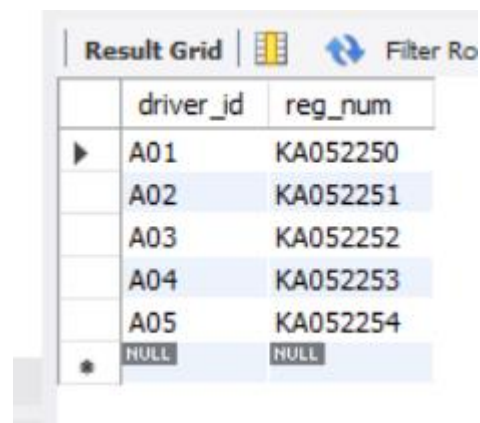
2. CAR

Result Grid			
	reg_num	model	year
▶	KA052250	indica	1990
	KA052251	toyota	1957
	KA052252	honda	1998
	KA052253	audi	2008
	KA052254	lambo	2005
*	NULL	NULL	NULL

3 ACCIDENT

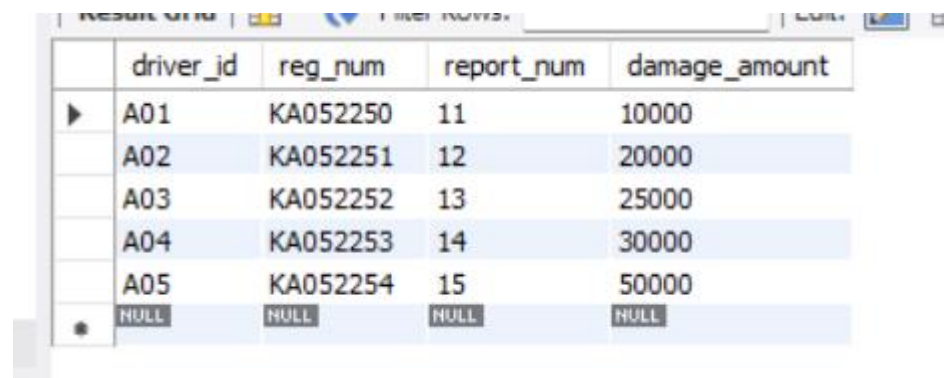
Result Grid			
	report_num	accident_date	location
▶	11	2003-01-01	Mysore Road
	12	2004-02-02	South end Circle
	13	2003-01-21	Bull temple Road
	14	2008-02-17	Mysore Road
	15	2005-03-04	Kanakpura Road
*	NULL	NULL	NULL

4 OWNS



	driver_id	reg_num
▶	A01	KA052250
	A02	KA052251
	A03	KA052252
	A04	KA052253
	A05	KA052254
•	NULL	NULL

5 PARTICIPATED



	driver_id	reg_num	report_num	damage_amount
▶	A01	KA052250	11	10000
	A02	KA052251	12	20000
	A03	KA052252	13	25000
	A04	KA052253	14	30000
	A05	KA052254	15	50000
•	NULL	NULL	NULL	NULL

EXECUTING QUERIES

1. UPDATE

```
74
75 • update participated set damage_amount=25000 where report_num=12;
76 • insert into accident(report_num, accident_date, location)
77 values (16, '2008-03-15', 'Domlur');
78 • select * from participated;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Cc
driver_id	reg_num	report_num	damage_amount	
A01	KA052250	11	10000	
A02	KA052251	12	25000	
A03	KA052252	13	25000	
A04	KA052253	14	30000	
A05	KA052254	15	50000	
NULL	NULL	NULL	NULL	

2. ORDERING BY YEAR (ASCENDING)

```
80
81 • select * from car order by year asc;
```

Result Grid	Filter Rows:	Edit:
reg_num	model	year
KA052251	toyota	1957
KA052250	indica	1990
KA052252	honda	1998
KA052254	lambo	2005
KA052253	audi	2008
NULL	NULL	NULL

3. NUMBER OF ACCIDENTS CAUSED BY SPECIFIC CAR

```

83 • select count(*) CNT from car c, participated p where c.reg_num=p.reg_num and model="audi";
84

```

Result Grid

	CNT
▶	1

4. TOTAL NO OF PEOPLE THAT MET ACCIDENT IN 2008

```

85 • select count( driver_id) CNT
86 from participated a, accident b
87 where a.report_num=b.report_num and year(b.accident_date)=2003;

```

Result Grid

	CNT
▶	2

5. PARTICIPATED IN DECENDING

```

89 • select * from participated order by damage_amount desc;
90

```

Result Grid

	driver_id	reg_num	report_num	damage_amount
▶	A05	KA052254	15	50000
	A04	KA052253	14	30000
	A02	KA052251	12	25000
	A03	KA052252	13	25000
	A01	KA052250	11	10000
*	NULL	NULL	NULL	NULL

6. Show all accidents (date and location)

90

91 • **select** accident_date, location **from** accident;

92

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	accident_date	location
▶	2003-01-01	Mysore Road
	2004-02-02	South end Circle
	2003-01-21	Bull temple Road
	2008-02-17	Mysore Road
	2005-03-04	Kanakpura Road
	2008-03-15	Domlur

7. Find drivers who caused damage ≥ 25000

92

93 • **SELECT DISTINCT** p.driver_id, p.name

94 **FROM** person p

95 **JOIN** participated par

96 **ON** par.driver_id = p.driver_id

97 **WHERE** par.damage_amount $\geq$ 25000

98

Result Grid



Filter Rows:

Export:



Wrap

	driver_id	name
▶	A02	pradeep
	A03	smith
	A04	venu
	A05	john

8. List each driver with the cars they own

```

145 • select p.name ,c.model
146      from person p
147      join owns o on o.driver_id=p.driver_id
148      join car c on c.reg_num=o.reg_num;

```

Result Grid	Filter Rows:	Export:	Wrap Cell C
	name	model	
▶	richard	indica	
	pradeep	toyota	
	smith	honda	
	venu	audi	
	john	lambo	

9. Show accidents and the drivers involved (including damage amount)

```

150 • use test;
151 • SELECT a.report_num,
152          a.accident_date,
153          a.location,
154          p.name AS driver_name,
155          pa.damage_amount
156      FROM accident a
157      JOIN participated pa ON a.report_num = pa.report_num
158      JOIN person p ON pa.driver_id = p.driver_id
159      ORDER BY a.report_num;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	report_num	accident_date	location	driver_name	damage_amount
▶	11	2003-01-01	Mysore Road	richard	10000
	12	2004-02-02	South end Circle	pradeep	25000
	13	2003-01-21	Bull temple Road	smith	25000
	14	2008-02-17	Mysore Road	venu	30000
	15	2005-03-04	Kanakpura Road	john	50000

10. Cars that never had an accident (owned but not in participated)

```
170 • SELECT c.reg_num, c.model, c.year
171 FROM car c
172 WHERE c.reg_num NOT IN (
173     SELECT DISTINCT reg_num
174     FROM participated
175 );
176
```

Result Grid

	reg_num	model	year
*	NULL	NULL	NULL

11. Latest accident (most recent accident_date)

```
177 • SELECT *
178 FROM accident
179 WHERE accident_date = (
180     SELECT MAX(accident_date)
181     FROM accident
182 );
183
184
```

Result Grid

	report_num	accident_date	location
▶	16	2008-03-15	Domlur
*	NULL	NULL	NULL

12. Average damage amount per driver

```

184 • SELECT driver_id,AVG(damage_amount) AS avg_damage
185 FROM participated
186 GROUP BY driver_id;
187

```

Result Grid |  Filter Rows: | Export:  Wrap Cell Content

	driver_id	avg_damage
▶	A01	10000.0000
	A02	25000.0000
	A03	25000.0000
	A04	30000.0000
	A05	50000.0000

13. Update: set damage_amount = 25000 for a specific car & report (example)

```

38 • UPDATE participated
39 SET damage_amount = 25000
40 WHERE reg_num = 'KA01AB1234'
41 AND report_num = 105;
42

```

14. Find drivers who caused the maximum damage in any single accident

```

193 • SELECT p.driver_id, p.name, pa.damage_amount
194 FROM participated pa
195 JOIN person p ON p.driver_id = pa.driver_id
196 WHERE pa.damage_amount = (
197     SELECT MAX(damage_amount)
198     FROM participated
199 );
200
201

```

Result Grid   Filter Rows: Export:  Wrap

	driver_id	name	damage_amount
▶	A05	john	50000

15. Show cars (model) involved in accidents with total damage > 20000

```
201 • SELECT c.model, SUM(pa.damage_amount) AS total_damage
202 FROM participated pa
203 JOIN car c ON pa.reg_num = c.reg_num
204 GROUP BY c.model
205 HAVING SUM(pa.damage_amount) > 20000;
206
207
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Conte

	model	total_damage
▶	toyota	25000
	honda	25000
	audi	30000
	lambo	50000

16. Create a view summarizing accidents with participants count and total damage

```

208 • SELECT a.report_num,
209         a.accident_date,
210         COUNT(pa.driver_id) AS participants_count,
211         SUM(pa.damage_amount) AS total_damage
212 FROM accident a
213 JOIN participated pa ON a.report_num = pa.report_num
214 GROUP BY a.report_num, a.accident_date;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content:				
	report_num	accident_date	participants_count	total_damage
▶	11	2003-01-01	1	10000
	12	2004-02-02	1	25000
	13	2003-01-21	1	25000
	14	2008-02-17	1	30000
	15	2005-03-04	1	50000